Hashdex

Crypto Investment Outlook

Opportunity meets Optimism

2024

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Getting back to ***boldness***

Marcelo Sampaio, CEO

What a difference a year makes.

Twelve months ago, the crypto industry was facing an unprecedented moment of distrust, bitcoin was trading around $16,000, and the cover of The Economist proclaimed “Crypto’s Downfall.”

Today, the fraudster at the heart of last year’s distrust faces over 100 years in prison, bitcoin is trading above $38,000, and BlackRock’s CEO believes crypto is playing a “flight to quality” role for investors.

As the dust has settled this year, both Bitcoin and Ethereum have been shining bright. The world’s first cryptocurrency was tested in the first quarter as US bank failures dominated headlines, but its sustained outperformance is a clear sign its role as a digital store of value in uncertain times is only getting stronger. And the acceptance and development of Ethereum is helping investors fully recognize the important role crypto will play in portfolios. Soon, this will be much easier for US investors as they gain access to both Bitcoin and Ethereum through the familiar structure of an ETF.

Like Bitcoin and Ethereum, Hashdex is also returning to boldness. This year was one of significant growth for us. Building on our 2022 product launches, we expanded partnerships in Latin America as our ETFs were recognized as industry leaders in the US and Europe. We have so much more to come in 2024.

We believe the work we’ve done during the crypto winter has prepared us to deliver for our investors as the current bull market takes shape. While we are entering 2024 with clear eyes and a level head, the energy behind the investment case for crypto is undeniable.

To help investors better understand the current opportunity this emerging asset class is presenting, our team put together the **Hashdex 2024 Crypto Investment Outlook** with a focus on the most impactful insights going into the new year.

As we transition to 2024, Hashdex will continue to focus on what is most important for crypto investors — providing secure and regulated access to this generational investment opportunity.

We thank you for your continued trust and look forward to what’s sure to be a monumental year for crypto investors.

#BacktoBoldness

Marcelo

December 4, 2023

Crypto’s new regime: Our 2024 investment themes

Samir Kerbage, CIO

Those of us that have been proselytizing crypto for many years have become accustomed to a roller coaster ride of highs and lows. But through this journey the most committed players in this space have not wavered from one belief: crypto ***will*** change the world for the better.

While extended crypto winters make this message difficult to deliver, down markets also help reaffirm our long-term commitment to an emerging asset class that has become more important with each market cycle.

As the crypto winter thawed this year and more signs of the adoption and acceptance of this asset class appeared, we were reminded that through its volatility, scandals, and self-interested fraudsters, the technologies powering crypto asset performance are aimed at improving the way individuals transact, create, and communicate.

This year, this asset class has become an even more important part of conversations around economic security, hedging geopolitical and monetary risk, and equitable access to the

financial system. Bitcoin (BTC), in particular, showed its tremendous value during a massive bank crisis, ongoing economic uncertainty, geopolitical stress, and war. As a result, crypto today has more impact within the global economy than at any other point in its 15-year history.

So, what does this new role for crypto mean for investors?

In short, it means we are at an unprecedented inflection point — a generational investment opportunity that will be fleeting as more institutions enter the space and network effects lead to exponential growth. The prospects for spot crypto ETFs in the US are as high as ever, and as regulated investment vehicles become more common, the role of crypto in portfolios will continue to expand.

As we enter the final days of 2023, it's clear that crypto investing is entering a new regime. To help investors understand this environment, we've identified three themes for 2024:

Crypto's new regime in 2024

Key milestones from 2023

1. Entering the ocean of "TradFi."

Large investors and corporations are adopting crypto at an enhanced pace, aligning with market conditions for an "institutional phase" as crypto becomes increasingly integrated with traditional financial systems.

- **Bitcoin** - ETF filings, role as "digital gold," proving to be both a risk-on and risk-off asset
- **Ethereum** - ETF filings, tokenization trend

2. Harnessing technological advances.

The underlying technologies driving the use of major crypto assets continue to develop and advance, helping scale networks and potentially reach billions of new users.

- **Bitcoin** - Utility (Inscriptions, BRC-20 tokens, Lightning Network developments)
- **Ethereum** - Improving scalability solutions

3. Benefiting from macro tailwinds.

A spot bitcoin ETF will greatly broaden investors' acceptance of this asset class and create new product innovation, supported by favorable market conditions, regulatory clarity, and the forthcoming halving.

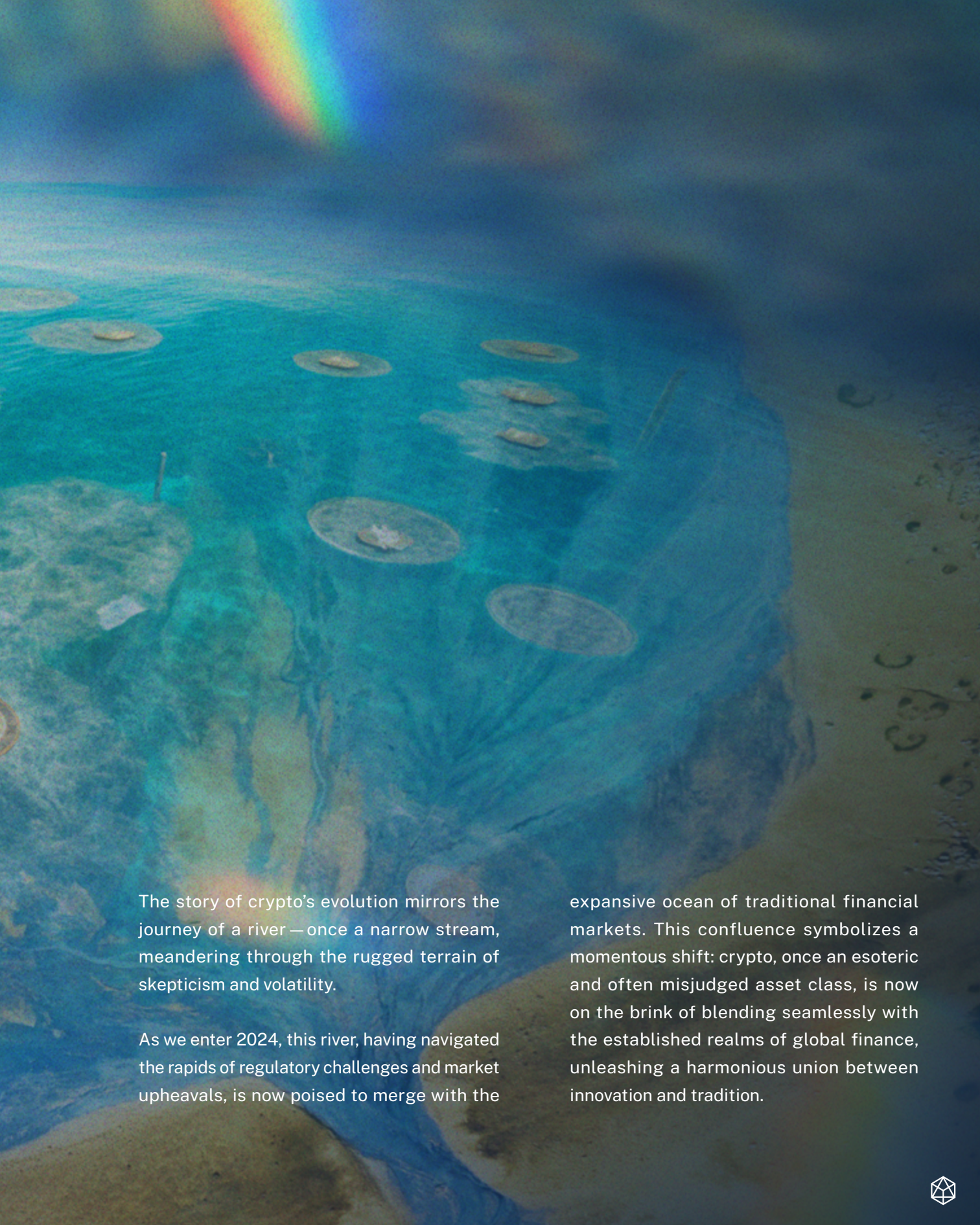
- **Bitcoin** - ETF filings, completion of the regulatory "trifecta"
- **Ethereum** - ETF filings, more favorable macro conditions, far less leverage in the ecosystem

Following in this report are brief articles that support these three themes and the underlying investment case for crypto in the new year.

We hope you find these insights useful.
Samir

Crypto's river meets the ocean of traditional finance

Pedro Lapenta, Head of Research



The story of crypto's evolution mirrors the journey of a river—once a narrow stream, meandering through the rugged terrain of skepticism and volatility.

As we enter 2024, this river, having navigated the rapids of regulatory challenges and market upheavals, is now poised to merge with the expansive ocean of traditional financial markets. This confluence symbolizes a momentous shift: crypto, once an esoteric and often misjudged asset class, is now on the brink of blending seamlessly with the established realms of global finance, unleashing a harmonious union between innovation and tradition.

2023: Rising from challenges to opportunities

After the big shock of the FTX collapse at the end of 2022, many thought crypto assets were dead. But in our 2023 Outlook we pinpointed many signals within the noise that, in our opinion, were not included in the public narrative.

Ultimately, 2023 has turned out to be a year of comeback. The worst is behind us and many positive changes emerged in 2023, including:

Market recovery:

Bitcoin regained $37,000 ($720 billion market cap) and ether is back above $2,000 ($245 billion)[1]. The total market cap (including stablecoins) is $1.4 trillion.

Role as a diversifier:

Crypto regained its place as an important part of a diverse portfolio, showing diminishing correlations with other major traditional assets such as the S&P 500, the Nasdaq 100, and even gold.

1 All prices and market capitalizations as of November 14, 2023.

Growth in user base:

The number of crypto users increased to 420 million globally, according to a study by Coinbase, with significant growth in the US — the largest market in the world.

Technological developments:

Major networks have gone through new use case developments and technological upgrades to tackle scalability and are now on a path toward several hundred million (perhaps billions) of new users. Tokenization, NFTs, and increasing interest in stablecoins by payment giants and traditional banks is driving this as well.

Traditional institutions' engagement:

Many global companies started using or integrating crypto and blockchain technology, including Deutsche Bank, Visa, Mastercard, JP Morgan, BlackRock, Fidelity, and Google. Many others applied for custody licenses, integrating crypto networks into their payments rails, filing for bitcoin and ether ETFs and integrating blockchains into product offerings and services.

Central banks' interest:

Most central banks have projects related to distributed ledger technology (DLT), central bank digital currencies, and bridges with decentralized finance (DeFi). A notable example is the Central Bank of Brazil, which is currently running the pilot project for an Ethereum-inspired financial system called DREX — an unquestionable sign of institutional support by a G20 country with more than 200 million people.

Regulatory advances:

83% of the G20 have either passed national crypto legislation or have crypto regulations in place. MiCA, the unified framework for crypto in the European Union, establishes clear rules for digital assets and service providers across its 27 member states.

And finally, **Bitcoin showcased its multifaceted behavior and reassured its resilience in the midst of diverse and adverse scenarios.** Throughout 2023, bitcoin displayed a remarkable ability to respond to several global events, acting as both a **risk-on** and **risk-off** asset:

- **Risk-on | Improving economic outlook:** In the first quarter, bitcoin responded positively to a risk-on market sentiment, showing its potential as a viable investment when decreasing inflation fueled expectations for more optimistic economic times.
- **Risk-off | Hedge against counterparty risk:** As the US banking crisis unfolded, led by the Silicon Valley Bank's collapse in early March, bitcoin emerged as a safe haven, reflecting its growing role as a protection asset in times of financial turmoil.
- **Risk-on | Anticipation of spot ETF approval:** The crypto market reacted favorably to news and developments around the spot bitcoin ETF applications by major institutions in the US, indicating investor optimism and reassuring confidence in increasing crypto institutional adoption.
- **Risk-off | Global conflicts:** Bitcoin outperformed in the wake of the Hamas attack on Israel, further underscoring its evolving role alongside gold and bonds as a "flight-to-quality" asset.

"I believe crypto will play that type of role, as a flight to quality."

BlackRock CEO Larry Fink in an October 16, 2023 Fox Business interview.

2024: A promising horizon

Crypto seems to be heading for another significant and transformative year. As the second semester of 2023 kicked in, we emphasized what we believe to be a generational opportunity, a rare moment in history when major secular drivers converge with an appealing cyclical, short-term entry point. As 2023 ends, those major drivers continue to be present, with leading 2024 factors being:

Rising demand for bitcoin:

The expected approval of a bitcoin ETF will bring a flush of fresh investments into the crypto markets.

Decrease in bitcoin issuance:

The upcoming halving (expected in late April) will reduce the amount of newly issued bitcoin. These events have preceded considerable bitcoin price appreciation in the past.

Favorable economic conditions:

The expected reduction in interest rates in the US and overall global liquidity improvement are likely to create a supportive environment for crypto assets in general.

These forces are no small things. A new spot bitcoin ETF will certainly attract billions of new dollars into the markets in the coming years, renovating and igniting a new bull cycle for crypto. On top of that, an approval by the SEC would cement a big win for the crypto industry and probably trigger new positive momentum in regulation in the largest capital market in the world. And to seal the deal, an approval of an ETF by firms such as BlackRock and Fidelity is the biggest stamp of approval and institutional backing this industry could get. The combination of these factors suggest interesting times ahead.

But after an impressive rally in 2023, are crypto markets overheated?

Seasoned investors can argue that crypto prices have already gone up enough and — because markets anticipate — the 2023 rally has already priced these factors in.

The most recent rally may be over excitement over potential crypto ETFs. If markets continue rallying in the weeks ahead, once the listings occur it's likely that market participants trading the event will proceed and take some profits off the table. On the flipside, if the ETF applications are delayed or denied altogether, some sell pressure is to be expected, as markets revert gains from the expectation of an approval.

But irrespective of any short-term price headwinds, our view is that we are currently in a crypto spring, preparing for a hot summer in 2024. Looking back at previous crypto summers, taking into consideration the caliber of institutional inflows to be expected, and adding up the risk-on and risk-off characteristics bitcoin exhibited in 2023, we expect major positive price movements in 2024 for the whole crypto industry.

The bottom line: An ocean of opportunity

As we venture into 2024, the crypto market's transition from river to ocean is not only a singular moment in this industry's history, but in the whole history of finance. Reflecting on the past years, we see a journey of resilience and growth: 2022's winter as a reversion to the mean and a flush out of bad industry players, followed by 2023's spring and the beginning of a recovery phase alongside the consolidation of innovation and rising institutional interest. In the crypto summer of 2024, these forces will coalesce in a transformative era in which crypto assets finally get integrated into mainstream traditional finance.

The challenges we faced in the past 24 months, particularly the FTX debacle, were not a death blow to crypto, but rather financial fraud, plain and simple. In fact, while negative price action was surely a short-term consequence, the aftermath of that event underscored the industry's robustness and potential, with its ability to recover from adverse events triggered by players whose businesses cannot stand the test of time. With continued adoption, regulatory progress, and heightened interest from major financial institutions, the crypto market is poised for a future rich with innovation and growth. The journey thus far has been remarkable, setting the stage for an even more vibrant and transformative future — the institutional phase of crypto.

The knock-on effects of a spot bitcoin ETF

Dramane Meite, CFA, Head of Product, US & Europe

The exact timing of a spot bitcoin ETF in the US remains unclear, but in 2023 the narrative around this product switched from a question of "if" to a matter of "when." We believe US investors will have access to a spot bitcoin ETF by the second quarter of the new year and that a spot ether ETF is likely to follow.

We've talked a lot over the years about why this product in this market matters, but it's worth revisiting how we think this will impact investors and the crypto ecosystem as a whole.

First, it's important to remember that crypto ETFs in the US will be a story of scaled distribution.

Spot ETFs will unlock the US retail and wealth opportunity — across financial advisors, direct retail, and private banks — a $50 trillion AUM market[2]. This is larger than Europe, Canada, and Brazil combined — three markets where crypto ETPs already exist.

And not only will distribution be scaled in the largest market in the world, but there will be captive distribution as well. For the first time, legacy asset managers with thousands of staff and trusted brands will offer their customers a crypto product. Four of the largest asset managers in the world, with close to $17 trillion in AUM, have filed for spot bitcoin ETFs.

This is creating exceptional conditions for the broad acceptance of crypto in distribution channels and among financial advisors, helping crypto to go mainstream as it is deployed into existing trusted channels and model portfolios.

2 Bloomberg Intelligence: $30 trillion financial advisor market, $12 trillion in direct retail, and $8 trillion in private bank assets.

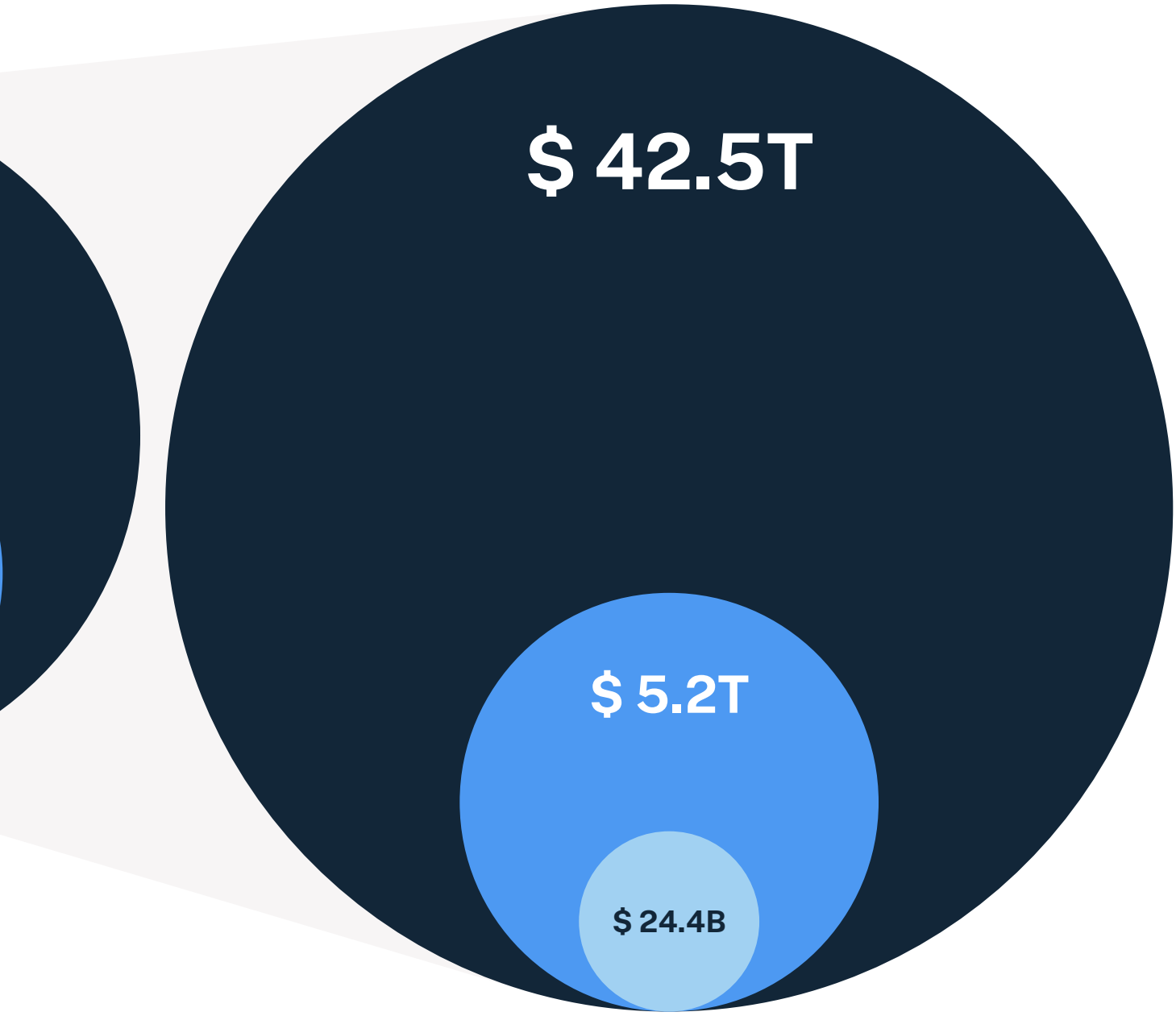
Product innovation will accelerate

The total global AUM of crypto ETF/ETP (including US Trusts) is currently $41 billion[3], but demand is estimated to grow to $600 billion.[4] How will this happen? We believe the developing bull market is creating an inflection point that will open up crypto ETPs to traditional investors.

Source: Hashdex analysis using data from BCG, ETF Book, ETF World, Fitch Ratings, Mordor Intelligence, and Oliver Wyman.
Data as of December 2022, circles drawn for ease of visualization and not in proportion to their true size.

3 Hashdex Analysis, as of November 15, 2023

4 Bernstein report referenced in September 29, 2023 Cointelegraph article.



US

0.4%

Once bitcoin and ether ETFs are available in the US, the conversation will quickly move to customized products, with risk and volatility to fit specific investor preferences, as well as what's next in the single-asset space. The latter will largely be driven by clarity from regulators and Congress. We expect most of the client interest in single-asset ETPs to be driven toward BTC and ETH, given their name recognition and little differentiation among incumbents.

As a result, one consequence of spot ETFs in the US will be to accelerate the shift quickly to beta and smart beta ETF solutions, in the US and globally, or hybrid products mixing crypto and tangential asset classes (e.g., gold, commodities, thematic equity). A few examples:

Beta (baskets) ETFs will become the next frontier, especially as staking will become more mainstream. Currently, Brazil is the only market where basket ETFs have a higher market share than single-asset ETFs. We expect to see a reversal of this as pent up demand for single-asset ETPs is met and financial advisors look to diversify exposure.

Thematic ETFs could be a winner this year, as the bull market that started with BTC moves to adjacent large cap sectors, and in particular smart contracts and their scaling solutions. A lot of work was done over the past 18 months but not reflected in price action yet. And the smart contract thesis is further boosted with tokenization, stablecoins, and expanding DeFi to real-world use cases.

Smart beta, such as momentum-based strategies, will likely make a comeback but with an emphasis on risk mitigation. There are also some interesting developments around applying quant and on-chain and fundamental analysis to building crypto portfolios. This could prove a winning combination as pure fundamental analysis can be tricky given the short history of many crypto assets.

Active strategies did poorly during the bear market as many proprietary and hedge fund-like strategies were taking counterparty risk to platforms such as FTX, 3AC, or undisclosed providers. It will take a while for that side of the market to get hot again, as investors will put a premium on transparency of the strategy and on understanding the returns on a risk-adjusted basis. There is, however, a strong case for active strategies and arbitrage strategies given the inefficiencies of crypto as an emerging asset class and its inherent volatility. However, due diligence has become more important than ever.

Now that we have moved beyond the crypto winter and are positioned for the next crypto summer, we'll continue to expect product innovation that meets the needs and demand from investors. As this evolution takes shape, investors can expect many more opportunities to put their capital to work in this emerging ecosystem of opportunities.

Bitcoin’s industrial age begins

Lucas Santana, Research Analyst

Several factors propelled the price of bitcoin[5] forward this year, including institutional acceptance, regulatory clarity, and its positive response to adverse events. One factor that has not been covered widely in the media but is driving demand for BTC is the arrival of new utility.

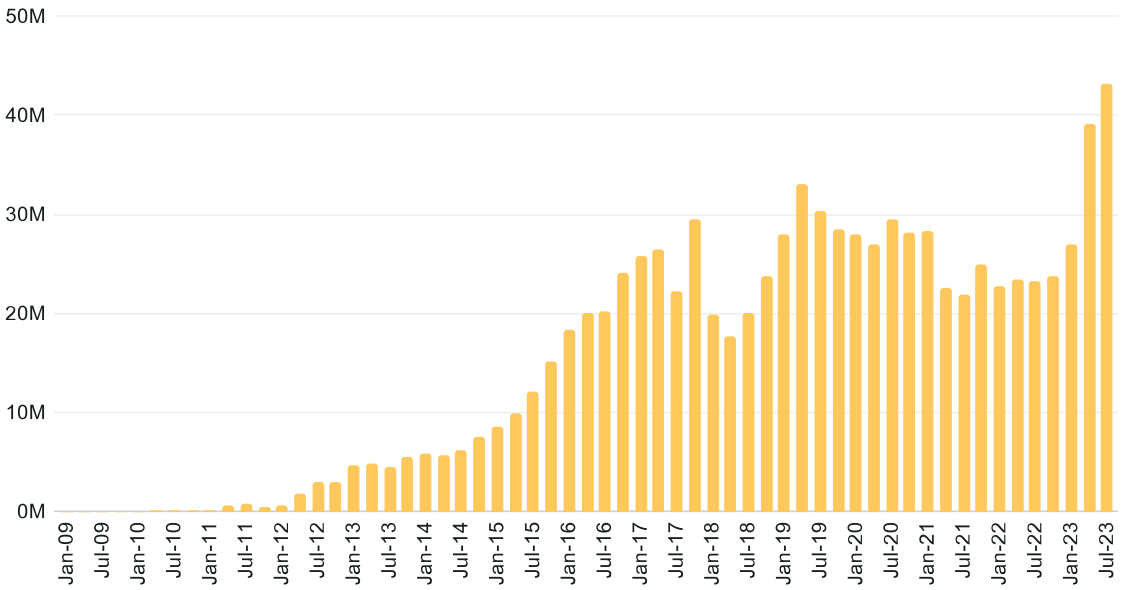
The year kicked off with the unexpected idea of BTC non-fungibility introduced by Ordinals, a set of rules that distinguishes the smallest fractions of a BTC[6] to inscribe data[7] on the Bitcoin blockchain. The process creates digital artifacts known as inscriptions, which enable NFTs[8] and fungible tokens[9] to be stored entirely on Bitcoin.

So far,[10] 40 million inscriptions among NFTs and fungible tokens have been created, a total of $96 million were paid to miners to register them on the Bitcoin blockchain, and inscriptions made Q3 2023 the first quarter where Bitcoin settled more than 40 million transactions in its nearly 15-year history. Later in the year, Lightning Labs — which supports the development of Bitcoin's major scaling solution called the Lightning Network[11] — released a tool to help bring stablecoins and other digital assets to Bitcoin's mainnet.[12]

5 Bitcoin with uppercase "B" stands for the network, whereas bitcoin with lowercase "b" refers to the asset.
6 A satoshi is the smallest fraction of a bitcoin and is equal to 0.00000001 BTC.
7 This is done via Taproot, the latest Bitcoin soft-fork activated in late 2021.
8 NFT is the acronym for non-fungible token, a unique digital asset stored on a blockchain.
9 That is, BRC-20 tokens, a specific inscription structure to issue fungible tokens.
10 As of November 16, 2023.
11 The Lightning Network is the leading second layer scaling solution on Bitcoin, allowing for BTC to be transferred in a fast and cheap manner without throttling the Bitcoin base layer.
12 Mainnet refers to the main (production) environment of a public blockchain network, rather than its testing environments.

The development of this utility is a sign that Bitcoin is re-emerging as a technological platform. These new uses and tools will serve as the building blocks that can be leveraged to create real use cases on the most secure and decentralized public blockchain.

Quarterly settled transactions on the Bitcoin network



Source: Hashdex Research (from January 3, 2009 to September 30, 2023).

For investors considering how this utility impacts the investment case for BTC, it's important to remember that more network activity inevitably drives demand for BTC. This demand, together with the maximum supply of BTC being set at 21 million and the halving scheduled for April 2024, is creating a unique and consequential moment for BTC investors.

As more network utility gains traction the investment case for bitcoin will expand beyond it being a store-of-value asset. As this happens, BTC can gradually take some market share of decentralized applications as it starts to tokenize traditional assets, potentially extending its total addressable market far beyond gold's $12 trillion in the years ahead. Bitcoin's "industrial age" as a programmability platform seems to just be getting started, and its potential implications to the 2024 investment case of the world's first crypto asset are very encouraging.

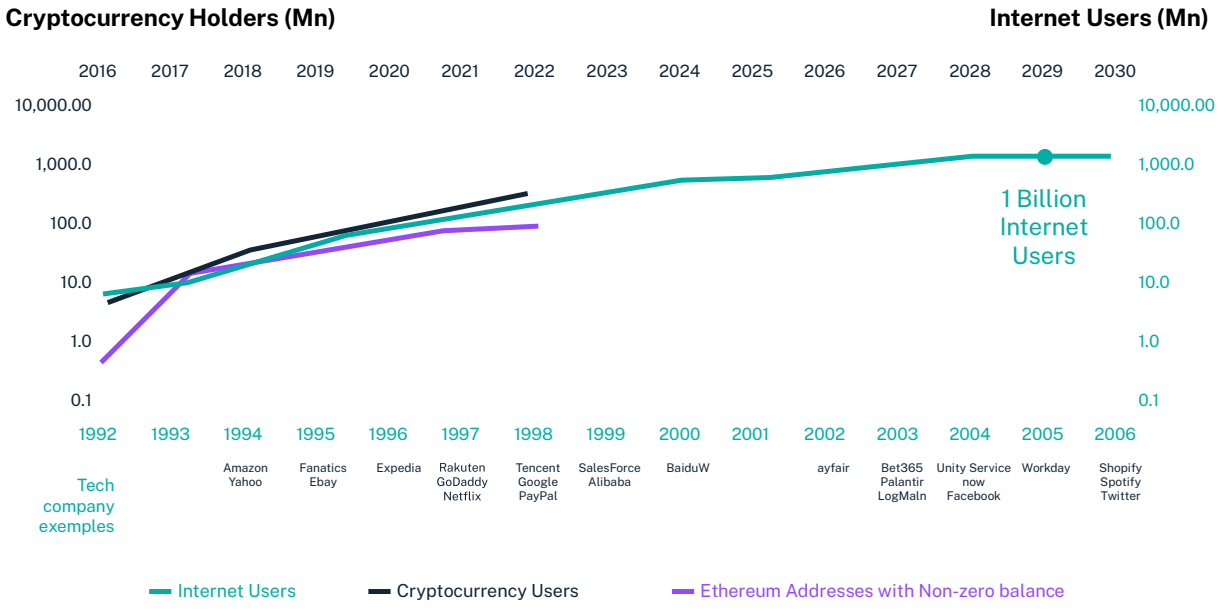
Ethereum propels a tokenized future

Yuri Alter Szaniecki, Research Analyst

While ether's (ETH) performance this year was modest relative to BTC, the fundamentals and technology remained strong. The new year may be the inflection point at which Ethereum shifts from a state of mere "promise" to acquiring the fundamental tools necessary to meet the demands of both institutions and the retail market.

As highlighted in our 2023 Outlook, Ethereum has faced challenges in terms of scalability. The network's limited throughput has historically posed a barrier to widespread adoption, hindering its ability to onboard users. These limitations matter because of the importance of users in determining ETH demand. However, there was notable progress in addressing these scalability issues in 2023 and further advancements are anticipated in 2024. These forthcoming solutions aim to facilitate seamless transactions across a diverse range of applications on the Ethereum blockchain, ensuring an efficient and unobstructed flow of data settlement.

Similar to the internet's transformative journey during its broadband[13] phase, Ethereum is currently poised to accumulate the necessary technological infrastructure to onboard users onto its diverse array of applications. We believe Ethereum's progression will continue to parallel the evolutionary trajectory of the internet, solidifying its position as a pioneering force in shaping the future of decentralized[14] digital transactions.

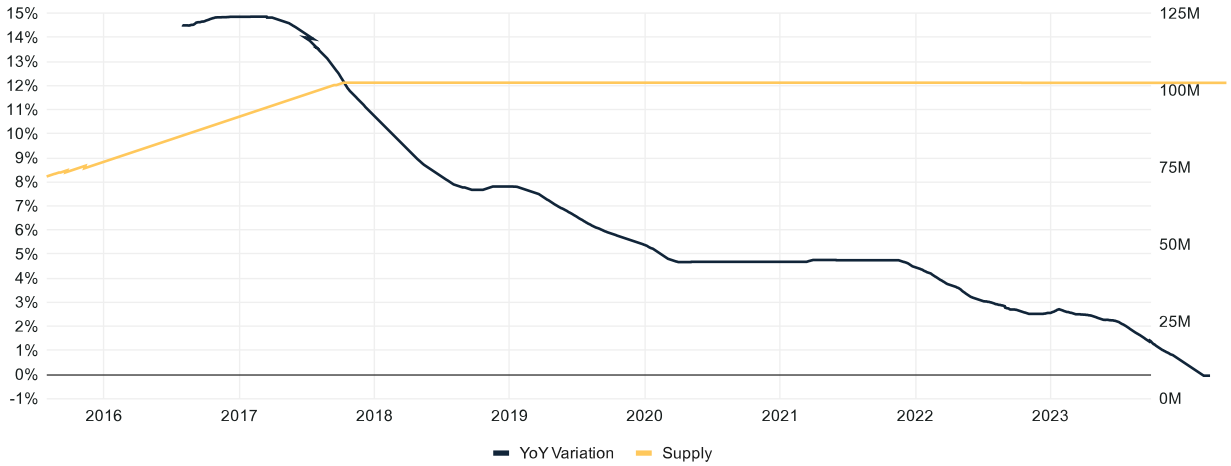


13 Broadband Internet access, often shortened to just broadband, is simply defined as "Internet access that is always on, and faster than the traditional dial-up access."
14 In blockchain, decentralization refers to the transfer of control and decision-making from a centralized entity (individual, organization, or group thereof) to a distributed network.

There are two other noteworthy trends for 2024. First is the tokenization of real-world assets.[15] Prominent banks like JPMorgan Chase and Deutsche Bank are embracing this innovative approach, and a recent report suggested traditional asset tokenization could exceed $16 trillion over the next 5-15 years. This is because the implementation of blockchain technology allows for a faster, more secure, and cost-effective solution for global transactions.

The second trend is how applications anticipated for development throughout the year may increase demand for network utilization and influence ETH demand. This is important to watch in light of the substantial reduction in Ethereum's inflation rate of newly minted ETH following the pivotal Merge update. Given the deflationary aspect of Ethereum's circulating supply throughout 2023, it is worth analyzing how the aforementioned exogenous demand shocks may affect prices in 2024.

Total supply and YoY supply variation of ether over time



Source: Hashdex Research with data from Etherscan and beaconcha.in (July 30, 2015 to Sept 30, 2023).

For investors evaluating ETH, they should also consider that Ethereum's transition from a Proof-of-Work consensus mechanism to Proof-of-Stake has made it more similar to a technology investment with measurable profitability. Valuation metrics like this that are being applied to Ethereum are helping boost its investment case and making it more familiar to traditional investors. In the new year, we anticipate Ethereum will benefit from this dynamic as it continues to drive tokenization as our economy becomes increasingly digitized.

15 Tokenization is the act of converting the value of a tangible or intangible asset into a token. The token itself is a piece of code made up of a distinctive asset reference, unique properties, and/or specific legal rights in accordance with the smart contract through which the token was generated. Real World Assets in crypto refers to the tokenization of tangible assets that exist in the physical world, that are brought on chain.

Brazil is a regulatory proof of concept

Julia Castelo Branco Rocha, Head of Legal and Compliance

Regulation was arguably the most important topic of 2023 in the crypto industry. There were many positive developments regarding clearer rules, including the approval of MiCA in Europe, proposed market structure and stablecoin bills in the US Congress, and Brazil's groundbreaking digital assets framework. But regulation will again be at the top of investors' minds in 2024. We think it's great to see Europe and the US move closer to regulatory clarity and believe the experience of crypto regulation in Brazil can inform what will happen in these other regions.

There has long been a recognition in Brazil that crypto assets should be treated like other financial instruments. The country's securities regulator (CVM) was early to appreciate the investor demand for this new asset class and so has been focused on investor protection. Brazil's regulators recognized quickly that there are millions of people investing in crypto on unregulated exchanges without high compliance standards. By creating a regulatory framework built by educated regulators you mitigate many of the risks associated with crypto investments, including liquidity and licensing, among others.

Brazil has become a proof of concept for crypto regulation. It was one of the first countries to approve a crypto ETF and it is one of the few to approve crypto basket ETFs. In October, Brazil's new framework for investment funds went into effect, which expressly includes crypto assets as a financial asset class similar to shares or notes.

This new framework for investment funds will undoubtedly revolutionize the investment funds industry, connecting traditional, centralized finance markets and decentralized finance (DeFi). This will help give investors confidence in holding this asset class in their respective investment portfolios, whether a retail investor, professional investor, or institution. The CVM has effectively eliminated all doubts about the possibility of Brazilian investment funds investing directly in crypto assets.

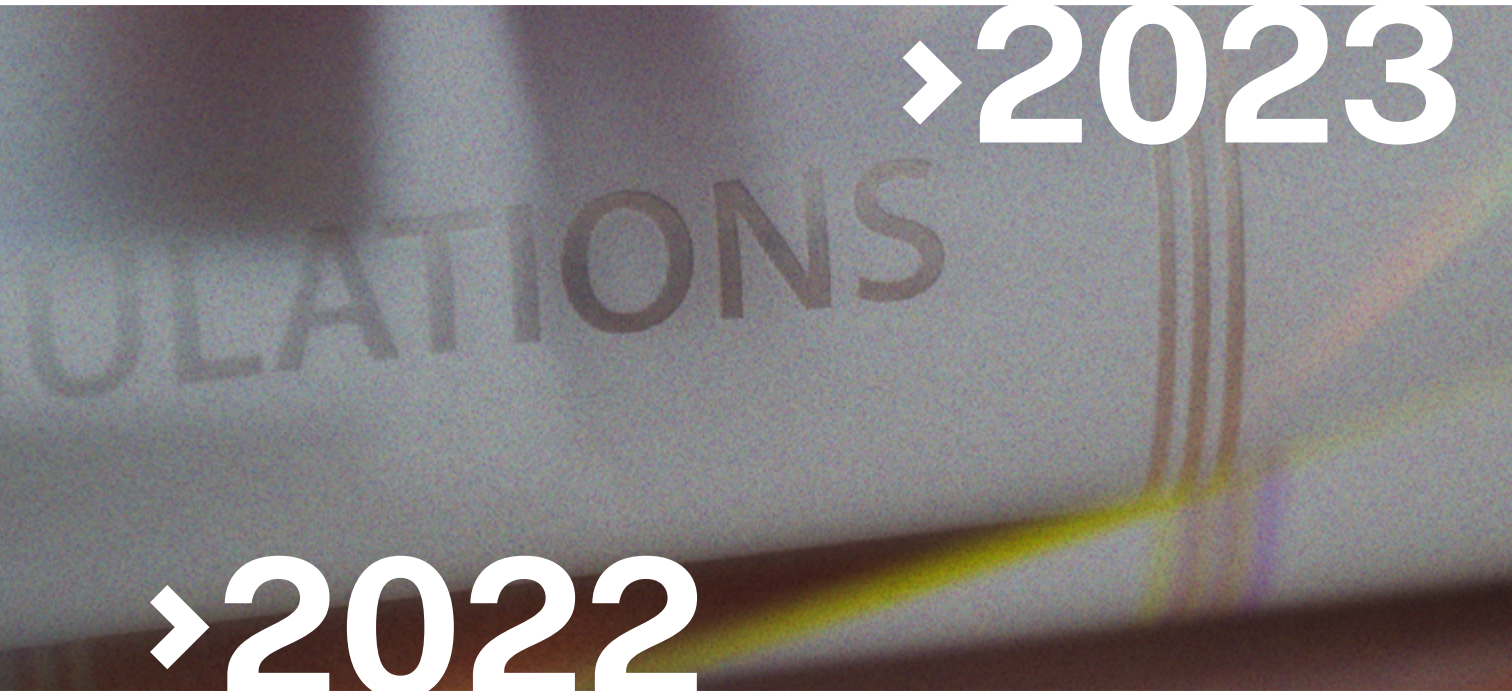
Timeline of Brazil's regulatory activity

January: The Brazilian Securities and Exchange Commission (CVM) indicates that several factors are being considered regarding crypto asset investments in Brazil.

September: CVM allows funds to invest indirectly in crypto assets and establishes several guidelines to be followed.

April: HASH11, the first crypto ETF in Brazil, begins trading on the Brazilian Stock Exchange (B3). The ETF tracks the Nasdaq Crypto Index, co-developed by Hashdex to track a basket of crypto assets.

June: The Central Bank of Brazil established that it is the body to regulate, authorize, and supervise providers of digital asset services.



October: CVM guidance clarifies how digital assets will be regulated in the market.

December: CVM establishes a new framework for investment funds, including digital assets as financial assets and allowing for the direct investment in crypto by investment funds.

December: The Cryptocurrencies Regulatory Framework becomes law, providing guidelines for the provision of services related to crypto assets, among other guidelines.

Will the US follow Brazil's lead on crypto asset regulation?

Certainly the US has unique considerations for a regulatory framework, but the basic building blocks — including ensuring investor protection — are the same.

But in the near term all eyes are on the spot bitcoin ETF in the US. There are many outstanding questions regarding how exactly a spot ETF will be approved, but 2023 was the year in which the market was finally given a signal from the SEC that a spot bitcoin ETF will be allowed. BlackRock's filing, the SEC losing its case against Grayscale, and the multiple applications for both spot bitcoin and ether ETFs are all strong signs that 2024 will be the year crypto ETFs arrive in the US.

We have met with the SEC to present our findings on the important topic of a spot ETF, and it's been widely reported that the SEC is meeting with other issuers as well. Ultimately, as we note throughout this report, the approval of a spot bitcoin ETF will be a massive regulatory hurdle cleared, permanently changing the investment landscape in the US.

Other countries will catch up

Brazil continues to advance in leaps and bounds, allowing access to this class by investors of all categories and in a regulated manner. Other markets, including Canada and Europe, are following the same path by recognizing crypto assets as financial assets that can be invested in by funds or other regulated vehicles. On the other hand, the US and China remain suspicious of crypto in general.

The pioneering spirit of Brazilian regulators and legislators has brought greater security and a more robust legal framework for crypto investors and service providers. We are optimistic that the US and other countries will take the opportunity in 2024 to build similar frameworks that protect investors and promote innovation.

U.S. SECURITIES AND EXCHANGE COMMISSION
MCMXXXIV

Optimism despite the past

João Marco Braga da Cunha, Head of Portfolio Management

"When I think about the future, I don't forget my past."

Paulinho da Viola

This line from a famous Brazilian song has been resonating in my mind as I reflect on this year and the year to come. There are two reasons for this.

The first one is contextual. It is impossible not to compare the current state of the crypto market with one year ago, when I was writing my contribution to our 2023 Outlook. The vast majority of our predictions were accurate and (thank God) the context now is completely different.

A year ago, both the Nasdaq Crypto Index (NCI) and bitcoin had lost over three quarters of their value in 12 months. Today,[16] they are at their highest values in 18 months. The worst days are far behind us and optimism is the dominant narrative.

As previously discussed, one of the main drivers of this appreciation cycle is the expectation of a spot bitcoin ETF approval in the US. Since June 15th, when BlackRock filed for its ETF with the SEC, bitcoin is up over 45%.[17] Along with the inexorable halving, the possibility of the SEC allowing the spot ETF is a main factor for confidence in 2024 — which takes me to the second reason I've been reflecting on this lyric.

The approval of a spot bitcoin ETF will be the final of three major US regulatory milestones for crypto as an asset class. The first two were the launch of bitcoin futures on CME in December 2017 and the listing of the first bitcoin futures ETF in October 2021. In both cases, these events were preceded by strong appreciation in crypto markets and new all-time highs were recorded. But then large drawdowns occurred in the subsequent 12 months.

It is, therefore, reasonable to ask if this drawdown will repeat. The straightforward answer is no, and it is worth taking a closer look at the two previous events in order to understand why. In spite of the similarities in terms of price action, the stories are different.

In 2017, when bitcoin futures were launched on CME, bitcoin was already up 500% year to date, driven by the ICO frenzy. Futures trading was like adding fuel to the fire. But even after the severe drawdown of approximately 70% from December 2017 to the end of June 2018, BTC was still up 490% for the 18-month period. All signs pointed to a classical price bubble burst.

The bitcoin futures ETF came into play in a completely different scenario. After a roughly sixfold surge from September 2020 to March 2021, BTC plunged mainly as a consequence of the crypto ban in China. By July, the drawdown surpassed 50%. In early August, the bitcoin futures ETF narrative gained momentum and drove prices up. The first ETF debuted on October 19, 2021 and a few days later bitcoin renewed its all-time high.

16 As of November 29, 2023
17 As of November 29, 2023

Neither the price action nor the frenzy were comparable to 2017. For instance, Google Trends data reveals that worldwide searches for Bitcoin in the fourth quarter of 2021 never reached a third of the peak from December 2017. Also, Bitcoin's on-chain activity metrics, such as the number transactions waiting for confirmation, weren't even close to the 2017 euphoria or the surge in early 2021. The conditions were different but once again a downturn followed the event, leading to another deep drawdown. This time, there was no bursting bubble to be blamed. As I wrote in an October 2022 article, over 70% of the price collapse was due to direct macro influence and idiosyncratic crypto events with highly leveraged players, such as Terra/LUNA, Celsius, and FTX.

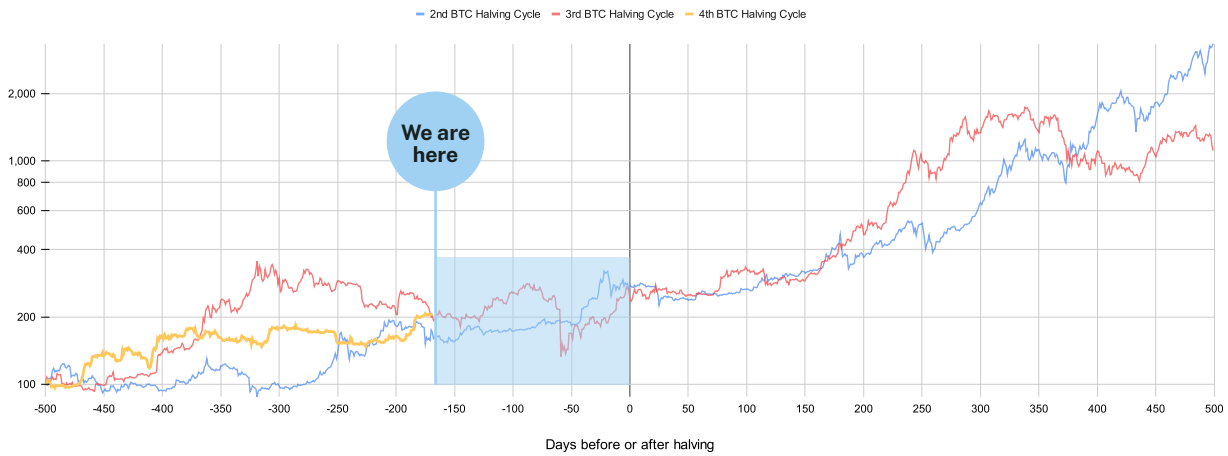
Crypto Regulatory Milestones

Signal	Event	Bitcoin Futures	Bitcoin Futures ETF	Spot Bitcoin ETF*	*Comments	Source
	Date	Dec 17, 2017	Oct 19, 2021	Jan 10, 2024	Expected date according to some analysts such as James Seyffart from Bloomberg	Bloomberg
Onchain	Bitcoin transactions waiting for confirmation	116,614	6,558	186,614	As of Nov 10, 2023	blockchain.com
	Bitcoin Halving in the next semester?	No	No	Yes	-	-
Price	Bitcoin returns 12 months before	2,158%	448%	112%	From Jan 10, 2023 to Nov 10, 2023	Bloomberg
	Bitcoin returns 12 months after	-82%	-70%	-	-	Bloomberg
Retail/ Institutional Interest	Google Trends (searches for Bitcoin/ month, 100 is the maximum)	100	37	20	As of Nov 10, 2023	GoogleTrends
	Bitcoin Futures Open Interest (USD billions)	N.A.	20	14	As of Nov 9, 2023	TheBlock
Macro	USA Core CPI 12 months after	2.2%	6.3%	2.7%	Analysts Expectation from Bloomberg	Bloomberg
	Fed Funds Rate change 12 months after	+0.8%	+3.0%	-0.8%	As of Nov 9, 2023	Bloomberg

So, why won't the aftermath of the spot bitcoin ETF look like the others? In terms of price, despite excellent performance in 2023, the prices of the main crypto assets are still far below all-time highs. Therefore, it is very hard to argue there is a bubble waiting to pop. The macro environment is much more normal than in 2022 when the Fed Funds Rate rose from near 0% to more than 4%, and the Core CPI was 5.7% YoY in December. In comparison, Fed Funds Futures implied interest rates are pointing to a cut of 80 basis points (bps) throughout 2024 and analysts anticipate Core CPI to register at 2.7% during that

period. That said, it is very unlikely that significant macro turmoil will show up to draw crypto prices down again. Finally, the overall leverage in the crypto industry is way smaller than two years ago, thanks to advances in regulation and natural selection (as most overleveraged players didn't survive). Idiosyncratic events with a meaningful impact on crypto prices have minimal probability.

In addition to these circumstances, we have the halving coming, possibly around the same time as a spot ETF approval. While the ETF can up shift the demand for bitcoin, the halving reduces the supply of new ones. The combination of both effects can be very powerful.



The crypto market teaches us to be humble and never dismiss the possibility of one or more black swans. Nevertheless, while we aren't forgetting the past, there are solid reasons to think very optimistically about 2024.

Glossary

CBDC
TAKING

BRC-20

BRC-20 is a token standard on the Bitcoin blockchain that utilizes Ordinals inscriptions to enable the creation and transfer of fungible tokens, providing functionalities for minting and transferring digital assets using Bitcoin.

CBDC

A central bank digital currency (CBDC) is a digital version of a country's fiat currency. Regulated by a country's monetary authority, CBDCs are designed to replace traditional fiat and increase ease of use for those that deploy them.

Inscription

Inscriptions refer to the metadata that is added or “inscribed” onto Bitcoin satoshis (sats), which are the smallest units of the bitcoin currency. These inscriptions can be attached to Bitcoin unspent transaction outputs (UTXOs) and store texts, images, audios, videos and even web applications.

Lightning Network

A scaling solution built atop Bitcoin that allows off-chain transactions with increased speed and lower transaction costs, without giving up security. Its capacity corresponds to the total BTC amount that all payment channels within the Lightning Network can send and receive.

Merge

Ethereum's transition from Proof-of-Work to Proof-of-Stake, which took place on September 15, 2022.

NFTs

Non-fungible tokens (NFTs) are a special type of token that represents a unique digital asset, such as a picture, a song or a video.

Proof-of-Stake

Consensus mechanism where validators lock an amount of native tokens as a guarantee of action in good faith, being randomly selected over time to propose and attest to new blocks of transactions.

Proof-of-Work

Consensus mechanism where miners utilize electricity and computer hardware to solve a cryptographic puzzle and become eligible to include a new block on the blockchain.

Stablecoin

A stablecoin is a digital currency created with the intent of holding a stable value with respect to another asset, such as fiat currencies, precious metals and other digital assets.

Staking

Staking involves pledging your cryptocurrency to support transaction validation on the blockchain. While you don't personally validate transactions, network computers handle this task, and major exchanges offer programs that make staking easy. The main motivation for staking is to earn rewards.

TradFi

Traditional finance (TradFi) refers to the financial system that exists apart from blockchain, cryptocurrency, and decentralized finance (DeFi), typically characterized by legacy financial institutions such as banks and large corporate financial enterprises that operate using a centralized model.

Tokeniza-tion

Tokenization is the act of converting the value of a tangible or intangible asset into a token. The token itself is a piece of code made up of a distinctive asset reference, unique properties, and/or specific legal rights in accordance with the smart contract through which the token was generated.

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